Section II

Wet-Chemical Processes

Chapter 3

Particle Deposition and Adhesion

Ahmed Busnaina[1], Kaveh Bakhtari[2], Jin-Goo Park[3]

[1]*Northeastern University, Boston, MA, United States;* [2]*Transient Electronics Inc., Brookline, MA, United States;* [3]*Hanyang University, Ansan, Republic of Korea*

Chapter Outline

3.1 Introduction to Particle Deposition and Adhesion 154
3.2 Particle Transport, Deposition, and Adhesion 154
3.2.1 Particle Deposition Mechanisms in Gas and Liquid Media 154
3.2.2 Drag and Lift Forces 154
3.2.3 Gravitational Force 158
3.2.4 Electrophoresis 158
3.2.5 Brownian Motion 159
3.2.6 Thermophoresis 160
3.2.7 The Double-Layer Electrostatic Force 163
3.2.8 Photophoresis 166
3.2.8.1 Radiometry 166
3.2.8.2 Radiation Pressure 166
3.2.9 Turbulence Effect 167
3.2.10 DLVO Theory 167
3.3 Particle Adhesion 168
3.3.1 van der Waals Force 168
3.3.2 Electrostatic Forces 168
3.3.2.1 Effect of pH 169
3.3.2.2 Effect of Ionic Strength 169
3.3.3 Short-Range Forces 171
3.3.4 Capillary Condensation 171
3.3.5 Effect of Deformation on Adhesion 174
3.3.6 Particle Adhesion Force Measurements 176
3.3.6.1 Adhesion Force Measurement by Atomic Force Microscopy 177
3.3.6.2 Measured Adhesion Forces in Liquid Media 178
3.4 Particle Removal 180
3.4.1 Megasonic Damage Mechanism 181
3.4.2 Damage-Free Megasonic Particle Removal 182
3.5 Summary 183
References 183

Handbook of Silicon Wafer Cleaning Technology. https://doi.org/10.1016/B978-0-323-51084-4.00003-4

3.1 INTRODUCTION TO PARTICLE DEPOSITION AND ADHESION

Surface contamination can result from particles deposited in the integrated circuit (IC) manufacturing environment and from particles generated by the manufacturing process or process tools. Improving the clean environment and controlling the cleanliness of process tools can solve only part of the contamination problem. Contaminant particles that may deposit on the wafer surface typically range in size from several hundred microns to less than 10 nm (nanometers). Types of particles and other contamination that reside on the wafer surface are discussed in this chapter. Surface particulate contamination is one of the primary reasons for yield problems in the IC industry. The adhesion forces of small particles are greatly affected by the processes that the substrate may be exposed to, such as thin-film deposition, polishing, etching, cleaning, rinsing, and drying steps.

The adhesion forces to be considered include van der Waals electrostatic forces and chemical bonds. Chemical bonds are usually orders of magnitude larger than the van der Waals force. As the size of the circuit line width decreases below 40 nm, contamination becomes more critical especially with respect to nanoscale particles at or below 40 nm. The removal of particles requires the understanding of all the forces involved plus their transport, deposition, and adhesion mechanisms. There is a need for efficient and reliable particle removal techniques capable of removing very small particles without causing surface damage. Later chapters discuss various types of particle removal techniques.

3.2 PARTICLE TRANSPORT, DEPOSITION, AND ADHESION

3.2.1 Particle Deposition Mechanisms in Gas and Liquid Media

During semiconductor processing and other high-precision manufacturing processes, particles may be deposited on a wafer or a substrate during manufacturing. Particle adhesion forces in different mediums will be discussed in this section.

3.2.2 Drag and Lift Forces

The drag force is one of the main forces that affect particle deposition and removal. The drag force is a function of velocity, viscosity, density, and the geometry and size of the particle and is the force exerted on the particle to remove it from the wafer surface. The coefficient of drag force, C_D, measured for a spherical particle immersed in a fluid is given by:

$$C_D = \frac{F_D}{1/2 p_l A U^2} \tag{3.2-1}$$

where F_D is the drag force, U is the velocity of the body relative to the medium, ρ_l is the density of the liquid, and

$$A = \frac{\pi D^2}{4} \tag{3.2-2}$$

is the cross-sectional area, A, of the sphere. As the diameter of a particle, D, approaches the mean free path of the fluid, the particle fluid no-slip boundary condition no longer holds, giving rise to slip conditions. The no-slip condition for viscous fluids assumes that at a solid boundary, the fluid will have zero velocity relative to the boundary. On the other hand, the slip condition or free slip condition assumes that the fluid velocity at the wall is equal to the free-stream fluid velocity (no velocity gradient or boundary layer).

The drag force, F_D, on a spherical particle in a Newtonian fluid can be expressed by the following equation:

$$F_D = \frac{C_D}{C_C}\frac{\pi}{8}\rho D^2 U^2 \tag{3.2-3}$$

where C_C is the Stokes—Cunningham slip correction and ρ is the density of the liquid.

Cunningham derived a correction factor for Stokes' law to account for the slip effect [1]. The expression for the Cunningham correction factor, C_C, is

$$C_C = 1 + 2\mathrm{Kn}(1.275 + 0.400)\mathrm{e}^{-0.550\mathrm{Kn}} \tag{3.2-4}$$

The Kundsen number, Kn, is used to describe the interaction between the particle and fluid. Kn is defined as:

$$\mathrm{Kn} = \frac{2\lambda}{D} \tag{3.2-5}$$

where λ is the fluid's mean free path; the average distance traveled by a molecule between successive collision. The drag force on a spherical object for a Reynold's number, Re_D, smaller than unity is as follows:

$$C_D = \frac{6\pi}{\mathrm{Re}_D} \tag{3.2-6}$$

$$F_D = 3\pi\mu UD \quad \text{for} \quad \mathrm{Re}_D \leq 1 \tag{3.2-7}$$

where μ is the dynamic viscosity of the medium. Eq. (3.2-7) is used for a suspended particle without any constraints around the particle when $\mathrm{Re} < 1$. Another case to analyze is when the spherical particle in a fluid is in contact with a surface. Because one side is constrained by the surface, the velocity profile is different than the previous case resulting in a different drag force on the particle.

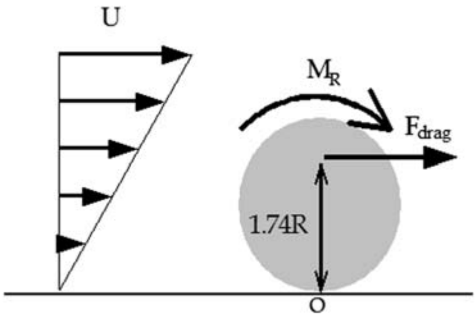

FIGURE 3.2-1 Schematic of the drag force, F_D, and removal moment, M_R, on a particle, with respect to the free streaming velocity, U, from Eq. (3.2-9). *Used with permission of the authors.*

O'Neill [2] showed that when a uniform linear shear flow, F_D, passes a sphere of diameter D on a surface, the drag force will be:

$$F_D = 16\mu D u_{D/2} \tag{3.2-8}$$

and the exact solution of the removal moment, M_R, about the contact point, shown in Fig. 3.2-1, is given by:

$$M_R = 1.74RF_D \tag{3.2-9}$$

where F_D is the drag force, R is the radius of the particle, μ is the dynamic viscosity of the media, U is the free stream velocity, and $u_{D/2}$ is the velocity of the fluid at the center of the particle.

The other hydrodynamic force on the particle is the lift force. The lift force results from the particle being present in a shear flow. In shear flow, adjacent layers of fluid move parallel to each other with different speeds. Viscous fluids resist this shearing motion. For a Newtonian fluid, the stress exerted by the fluid in resistance to the shear is proportional to the strain rate or sheer rate. Saffman [3] was the first to consider the problem of lift on a sphere translating in an unbounded linear shear flow field. Saffman's analysis applies to a sphere moving slowly in a strong shear field as shown in Fig. 3.2-2.

Defining the Reynold's number, Re_G, based on the velocity gradient, G, the sheer rate, F, at the wall, and kinematic viscosity υ as

$$\mathrm{Re}_G = \frac{FD^2}{\upsilon} \tag{3.2-10}$$

the restrictions on Saffman's analysis can be stated as:

$$\sqrt{\mathrm{Re}_G} \gg |\mathrm{Re}_P| \tag{3.2-11}$$

Re_P is the slip Reynolds number and Re_G is the sheer Reynold number. Saffman also assumed that both Re_G and Re_P are small compared with unity.

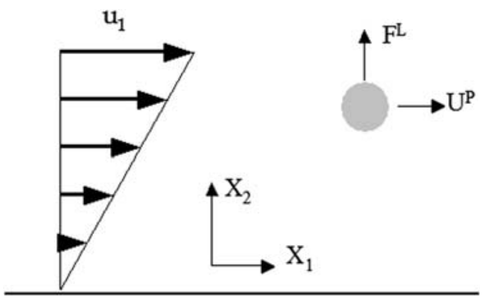

FIGURE 3.2-2 Schematic of the lift force in a shear flow in Cartesian coordinate system (X_1, X_2) [3]. F^L is the lift force on the particle, U^P is the particle velocity, and u_1 is the free stream velocity. *Reprinted from J. Fluid Mech., Cambridge University Press.*

The expression for the force of lift on a particle in a simple shear flow, F^L, is given as:

$$F^L = \frac{1.615\mu D^2}{\upsilon^{1/2}} \frac{du_1/dx_2}{|du_1/dx_2|^{1/2}} (u_1 - U^P) \tag{3.2-12}$$

where, u_1 is the fluid velocity, U^P is the particle velocity, du_1/dx_2 is the sheer rate.

McLaughlin [4] generalized Saffman's analysis by removing the restriction imposed by Eq. (3.2-13) and derived a general expression for the lift force. The parameter ε is defined as:

$$\varepsilon = \frac{\sqrt{\mathrm{Re}_G}}{\mathrm{Re}_P} \tag{3.2-13}$$

$\varepsilon \to \infty$ corresponds to the Saffman lift Eq. (3.2-14). The expression given by McLaughlin for the lift force, F^L is:

$$F^L = \frac{1.615\mu D^2}{\upsilon^{1/2}} \frac{du_1/dx_2}{|du_1/dx_2|^{1/2}} \left(u_1 - U^P\right) \left(\frac{J(\varepsilon)}{2.255}\right) \tag{3.2-14}$$

The function J depends on the magnitude of ε and is given by the approximation relations:

$$J(\varepsilon) = 2.255 - 0.6463/\varepsilon^2 \quad \text{for} \quad \varepsilon \gg 1 \tag{3.2-15}$$

$$J(\varepsilon) = -32\pi^2\varepsilon^5 \ln\left(1/\varepsilon^2\right) \quad \text{for} \quad \varepsilon \ll 1 \tag{3.2-16}$$

The function J can be considered to be a dimensionless lift force which has a value 2.255 as $\varepsilon \to \infty$. As the value of $|\varepsilon|$ decreases, J decreases rapidly.

3.2.3 Gravitational Force

In the presence of a gravitational field, a particle will experience a body force equal to the mass of the particle multiplied by the gravitational acceleration due to gravity, g_i. For aerosols, the buoyancy of the particle is negligible and therefore gravity is the only body force acting on the particle.

In liquids, the buoyancy is comparable with the gravitational force, g. Eq. (3.2-17) can be used to calculate the total body force, F_{Body} acting on the particle:

$$F_{\mathrm{Body}} = \frac{4}{3}\pi R^3 \left(\rho_p - \rho_m\right) g \tag{3.2-17}$$

where ρ_p and ρ_m are the density of the particle and medium, respectively, and R is the particle radius. It can be shown from Eq. (3.2-17) that if the particle density is in the same order as the density of the medium, the net body force on the particle will be zero. If the shear Reynolds number is sufficiently large, the lift force exceeds the gravitational force and the sphere separates from the wall.

3.2.4 Electrophoresis

Electrophoresis is the movement of an electrically charged substance under the influence of an electric field. Thus, particles can move in a fluid because of the charges that can occur on their surface. Electrophoresis movement is caused by the Lorentz force [5], which may be related to fundamental electrical properties of the body under study and the ambient electrical conditions by the equation given below:

$$F_E = Eq \tag{3.2-18}$$

where F_E is the electric force on the particle, E is the electric field intensity, and q is the net charge associated with the particle. In a constant and homogeneous electric field, a charged particle accelerates until the drag force, which is proportional to the velocity as shown in Eq. (3.2-3), equals the electrostatic force. When considering particle motion in a viscous medium, it is convenient to define the mobility, μ, of a particle as the limiting velocity attained under unit force:

$$\mu = v/E \tag{3.2-19}$$

where v is the limiting velocity also known as drift or terminal velocity and E is the electric field intensity.

The electrophoretic mobility, μ, can be expressed in general through the ζ potential as:

$$\mu = \frac{\varepsilon\xi}{\eta} f(\kappa R) \tag{3.2-20}$$

where η is the solvent viscosity, ε is the dielectric constant, κ^{-1} is the Debye screening length, R denotes the colloidal radius, and f is a function of salt concentration (as defined by the Debye length) and colloidal radius.

Smoluchowski gave the earliest solution to the problem in 1921 [6] for the case $\kappa R >> 1$, which is for a thin double layer, defined in Section 3.2.7. Smoluchowski concluded that the electrophoretic mobility should have the following form known as the Helmholtz–Smoluchowski equation.

$$\mu = \frac{2\varepsilon\xi}{\eta} \tag{3.2-21}$$

Hückel solved the electrophoresis problem for the opposite extreme condition of a very thick double layer ($\kappa R << 1$) in 1924 [7]. In this case, the field-lines are almost unaffected by the particle and Hückel equation for mobility is given as follows:

$$\mu = \frac{2\varepsilon\xi}{3\eta} \tag{3.2-22}$$

An electric field can be represented diagrammatically as a set of lines with arrows on, called electric field-lines, which fill space. The direction of the electric field is everywhere tangent to the field-lines. The magnitude of the field is proportional to the number of field-lines per unit area passing through a small surface normal to the lines.

3.2.5 Brownian Motion

Brownian motion is named after the Scottish botanist Robert Brown who was the first to study such random motion and fluctuations in 1827 [8]. If a number of particles are subjected to Brownian motion in a given medium, with no preferred direction for the random oscillations, then over a period of time the particles will spread evenly throughout the medium. Thus, if A and B are two adjacent regions and, at time t, A contains twice as many particles as B, at that instant the probability of a particle leaving A to enter B is twice as great as the probability that a particle will leave B to enter A. A physical process in which a substance tends to spread steadily from regions of high concentration to regions of lower concentration is called diffusion. Thus, diffusion is a macroscopic manifestation of Brownian motion on the microscopic level. It is thus possible to study diffusion by simulating the motion of a Brownian particle and computing its average behavior.

When a small particle is suspended in a fluid, it is subject to the collisions with the gas or liquid molecules. For fine particles, the instantaneous momentum (from the collision with molecules) on the particle causes the particles to move on a random path known as Brownian motion. The effects of Brownian motion could be included in the equation of motion as an additional force term. One approach for modeling Brownian force is to use a Gaussian white noise process with spectral intensity S_0 given by the equation [9]:

$$S_0 = \frac{216 \upsilon k_B T}{\pi^2 \rho D^5 \left(\rho_p / \rho\right)^2 C_C} \tag{3.2-23}$$

where T is the absolute temperature of the fluid, υ is the kinematic viscosity, k_B is the Boltzmann constant, ρ is the density of the fluid, ρ_p is the density of the particle, D is the particle diameter, and C_C is the Cunningham correction factor, defined in Eq. (3.2-1). The amplitude of the Brownian force, F_B, components is of the form:

$$F_B = Z\sqrt{\frac{\pi S_0}{\Delta t}} \tag{3.2-24}$$

where Δt is the time step and Z is the Gaussian random number bounded by -1 and $+1$, and S_0 is the spectral intensity. The amplitude of Brownian force components is evaluated at each time step. Depending on the application, the randomization effect of Brownian motion can also be analyzed as mean time of travel. From Einstein's equation [10], the mean time t_B taken for a particle to move a distance Δd in one dimension can be derived [11] by:

$$t_B = \frac{3\pi \mu D (\Delta d)^2}{k_B T} \tag{3.2-25}$$

where T is the temperature and μ is the mobility.

3.2.6 Thermophoresis

Thermophoresis is a drag force that arises from asymmetrical interaction of a particle with the surrounding gas molecules as a result of a temperature gradient. In simple terms, the particle will be repelled away from a hot surface and attracted to a cold surface. Tyndall first described the phenomenon in 1870 [12] when he observed a dust-free zone in a dusty gas around a hot body.

One of the earliest attempts to calculate the forces on spherical particles in a gas at rest with an existing temperature gradient was that by Epstein in 1929 [13]. He derived expressions for the thermophoretic force and velocity acquired by the particle in the slip flow regimen ($\mathrm{Kn} \leq 0$). Epstein's result is found to be in reasonable agreement with experimental results for $\mathrm{Kn} \leq 0$ and for particles with low thermal conductivity. However, it seriously underestimates the thermal force on particles of high conductivity.

Since Epstein's publication, a number of attempts have been made to improve the calculation to account for the discrepancies between the theoretical and experimental results for high—thermal conductivity particles and large Knudsen numbers. These improvements fall into the following four categories:

1. Hydrodynamic analysis based on the Navier–Stokes–Fourier theory, with slip corrected boundary conditions
2. Analysis based on higher order kinetic theory approximations to the continuum equations and boundary conditions
3. Analysis that employs phenomenological equations based on postulates of irreversible thermodynamics
4. Analysis based on the solution of Boltzmann equation

The hydrodynamic approach is the simplest and yields the most satisfactory result, which is explained in this section.

Brock [14] carried out the first hydrodynamic analysis and found the following relationship for the thermophoretic force for the near-continuum limit (Kn → 0):

$$F_i^{\mathrm{th}} = \frac{-12\pi\mu\upsilon R C_S \left(\dfrac{k_f}{g_p} + C_t \mathrm{Kn}\right) \dfrac{\nabla T}{T}}{(1 + 3C_m \mathrm{Kn})\left(1 + 2\dfrac{k_f}{k_p} + 2C_t \mathrm{Kn}\right)} \tag{3.2-26}$$

where F_i^{th} is the thermophoretic force on the particle in the 'i' direction, υ is the kinematic viscosity, μ is the mobility, T is the absolute temperature of the fluid, ∇T is the change in temperature across the gradient of the fluid, R is the radius of the particle, C_s is the thermal slip coefficient, C_t is the temperature jump coefficient, and C_m is the momentum exchange coefficient. All these coefficients are of the order of unity and must be obtained from kinetic theory. Brock's results initially were not in good agreement with experimental results for particles with high thermal conductivity or more precisely for Kn $\ll k_f/k_p$. k_p and k_f are defined as the thermal conductivities of the particle and fluid, respectively. The correction between Brock's theory and experiments was improved by adjusting the values of the coefficients. Modifications of the hydrodynamic theory since Brock's work have been mere improvements to the values of these coefficients.

The collisionless limit (Kn → ∞) for the thermophoretic force was given by Waldmann in 1966 [15] to be:

$$F_i^{\mathrm{th}} = -2\pi\mu\upsilon \frac{R^2 \nabla T}{\lambda T} \tag{3.2-27}$$

where λ is the fluid's mean free path. Examining the limit of Eq. (3.2-26) as Kn → ∞, with the exception for the multiplication factor (C_s/C_m), this equation is identical to Eq. (3.2-27). Batchelor and Shen [16] found this coefficient experimentally to be 1.0008. This means, only 0.08% error is involved by using Eq. (3.2-26). Therefore, Eq. (3.2-26) is a useful formula for calculating the thermophoresis force for the entire range of Knudsen numbers ($0 \leq \mathrm{Kn} < \infty$) and agrees reasonably with experiments using high−thermal conductivity particles.

Hinds, in 1982, presented the thermophoretic force on a particle of diameter D in terms of the fluid pressure, P, as [17]:

$$F_i^{\mathrm{th}} = \frac{-P\lambda D^2 \nabla T}{T} \quad \text{for} \quad D < \lambda \tag{3.2-28}$$

The mean free path λ is a function of pressure, P, and temperature, T, and can be expressed as:

$$\lambda = \frac{k_B T}{\sqrt{2\pi P D_m^2}} \tag{3.2-29}$$

where D_m is the molecular collision diameter. If the molecules were hard spheres, then D_m would be constant and have no dependence on temperature. However, because the molecules are not rigid spheres, the collision diameter is a function of temperature. Bird et al. in 1960 calculated D_m from the momentum transport coefficient to be [18]:

$$D_m = \left(\frac{2}{3\pi\mu} \sqrt{\frac{M k_B T}{1000\pi}} \right)^{1/2} \tag{3.2-30}$$

where M is the molecular mass of the medium.

The relationship between the thermophoretic force, particle diameter, D, and medium temperature, T, can be expressed as:

$$F_i^{\mathrm{th}} \propto D^2 \nabla T \mu T^{-1/2} \tag{3.2-31}$$

This shows that thermophoresis is independent of the pressure because $P\lambda$ is constant for a fixed temperature. To show the effect of thermophoresis on particle motion, it is appropriate to find the acceleration of the particle caused by the thermophoretic force. Therefore, dividing Eq. (3.2-31) by the particle mass leads to the following relationship:

$$f_i^{\mathrm{th}} \propto D^{-1} \nabla T \mu T^{-1/2} \tag{3.2-32}$$

where f_i^{th} is the thermophoretically induced particle acceleration. Eq. (3.2-32) indicates that the effect of thermophoresis is inversely proportional to the particle diameter; thus, smaller particles will have a higher induced particle acceleration. The effect is also inversely proportional to the square root of the temperature and directly proportional to the delta in temperature of the particle and the attracting body.

3.2.7 The Double-Layer Electrostatic Force

When an uncharged surface is immersed into a liquid, it will attain a surface charge because of preferential adsorption of ions present in the liquid or because of dissociation of its surface groups. The final surface charge is balanced by an equal but oppositely charged region of counterions surrounding the charged particle, some of which are bound to its surface within the so-called Stern layer, while others form the diffuse electric double layer, both layers are shown in Fig. 3.2-3. The Stern layer is the layer of counterions that attach to a charged surface. These ions are temporarily bound and screen the surface charge.

The shear plane, also known as the slipping plane, is an imaginary surface separating the thin layer of liquid bound to the solid surface, which shows elastic behavior as compared with the remainder of liquid, which shows normal viscous behavior. The electric potential at the shear plane is called the zeta potential.

The variation of electric potential, ψ, with distance from a charged surface of arbitrary shape is described by Poisson's equation:

$$\nabla \cdot \nabla \psi = \nabla^2 \psi = -\left(\frac{\rho_e}{\varepsilon_r \varepsilon_0}\right) \tag{3.2-33}$$

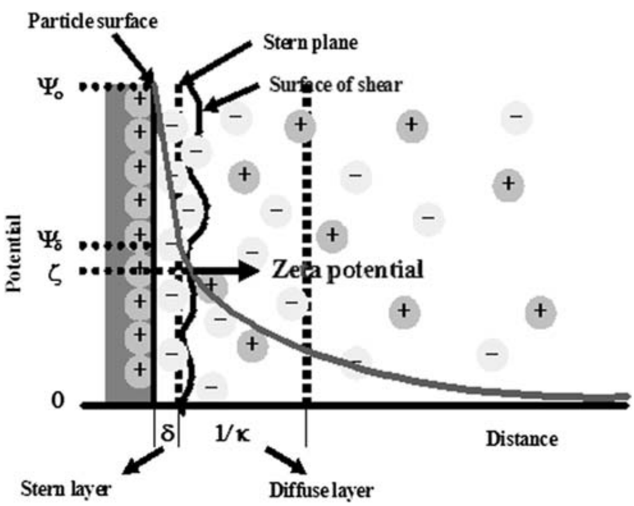

FIGURE 3.2-3 Schematic of zeta potential and the double layer. *Used with permission of the authors.*

where ρ_e is the local electrical charge density or ion concentration, ε_r is the dielectric constant (relative permittivity) of the fluid, and ε_0 is the permittivity of free space. The ion concentration follows the Boltzmann distribution:

$$\rho_e = \sum_i z_i e n_i = \sum_i z_i e n_i \infty \exp\left(-\frac{z_i e \psi}{k_B T}\right) \tag{3.2-34}$$

where k_B is the Boltzmann's constant, e is the electronic charge, T is the temperature, z_i is valency or charge, n_i is the number of ions of type i per unit volume, and $n_i\infty$ is the bulk concentration that is the concentration far from surface.

Combining Eqs. (3.2-33) and (3.2-34) gives the Poisson–Boltzmann equation:

$$\nabla^2 \psi = -(e) \sum_i z_i n_{i\infty} \exp\left(-\frac{z_i e \psi}{k_B T}\right) \tag{3.2-35}$$

Eq. (3.2-33) does not have an explicit general solution. Considering the situation for which $z_i e\psi/k_B T < 1$ ($\psi < 25$ mV at 25°C), a linearized Poisson–Boltzmann equation is obtained:

$$\nabla^2 \psi = k_B^2 \psi \tag{3.2-36}$$

$$\kappa^2 = \sum_i \frac{\rho \infty_i e^2 z_i^2}{\varepsilon \varepsilon_0 k_B T} \tag{3.2-37}$$

and the Debye length κ^{-1} is the characteristic thickness of diffuse electric double layer. Linearized Poisson–Boltzmann equation gives quite good results even for potential 50–60 mV [19]. Define $\rho\infty$, define ε (vs. ε_r, in Eq. 3.2-33)

Using spherical coordinates, linearized Poisson–Boltzmann equation becomes:

$$\frac{1}{r^2}\frac{d}{dr}\left(r^2 \frac{d\psi}{dr}\right) = k_B^2 \psi \tag{3.2-38}$$

Under the boundary conditions: $r = R$ (particle radius), $\psi \to \psi_0$; and $r \to \infty$ $\psi = 0$, the solution of this equation is given by:

$$\psi = \psi_0 \frac{R}{r} \exp[-\kappa(r - R)] \tag{3.2-39}$$

The overall electro-neutrality requires that the total charge of the diffuse double layer must be equal (and opposite) to the charge on the particle surface. If σ is the surface charge density, then the condition of electro-neutrality gives:

$$4\pi R^2 \sigma = -\int_R^\infty \rho \, 4\pi r^2 dr \tag{3.2-40}$$

From Eq. (3.2-39):

$$\rho = -\varepsilon_r \varepsilon_0 \frac{1}{r^2} \frac{d}{dr}\left(r^2 \frac{d\psi}{dr}\right) \tag{3.2-41}$$

and using the boundary condition: $r \to \infty$, $d\psi/dr = 0$, Eq. (3.2-40) is obtained:

$$\sigma = -\varepsilon_r \varepsilon_0 \left(\frac{d\psi}{dr}\right)_R \tag{3.2-42}$$

and using the expression for ψ given in Eq. (3.2-39), the following equation is obtained for the particle charge, q:

$$q = 4\pi R^2 \sigma = 4\pi R \varepsilon_r \varepsilon_0 (1 + \kappa R)\psi_0 \tag{3.2-43}$$

In reality, the surface potential ψ_0 cannot be measured. Within the Stern layer, the charge distribution is very complicated and the potential does not follow Eq. (3.2-39), only outside the Stern layer the diffuse double layer can be described by this equation. Zeta potential, ζ, is measured at the surface of shear, which is at the location roughly equivalent to the Stern surface.

The zeta potential will be used ζ instead of ψ_0 in Eq. (3.2-43), yielding:

$$q = 4\pi R^2 \sigma = 4\pi R \varepsilon_r \varepsilon_0 \,(1 + \kappa R)\xi \tag{3.2-44}$$

Another application for zeta potential is when two surfaces with such charges approach each other. In such case, the electrical double layer force which could be attractive or repulsive forms between them. An example is the double-layer force on a particle sitting on a substrate in the presence of a solution.

Analytic expressions for the forces and free energies of electrical double-layer interaction, even for spherical particles, are only available as approximate expressions. Each approximate expression is derived under certain assumptions and conditions. The Hogg—Healy—Fuerstenau (HHF) approximation [19], which is based on the one-dimensional linear Poisson—Boltzmann and Debye—Hückel equation, is commonly used. At small separation distances, the HHF model overestimation the force of adhesion. This equation calculated the double-layer force, $F(d)$, under the assumption of constant surface potential, ψ:

$$F(d) = 2\pi\varepsilon\varepsilon_0 \frac{R_1 R_1}{R_1 + R_2}\left(\Psi_{01}^2 + \Psi_{02}^2\right) \frac{\kappa e^{-\kappa d}}{1 - e^{-2\kappa d}} \left[\frac{2\Psi_{01}\Psi_{02}}{\Psi_{01}^2 + \Psi_{02}^2} - \kappa d\right] \tag{3.2-45}$$

where d is the separation distance between two spheres, ε is the dielectric constant of the medium, ε_0 is the dielectric permittivity of a vacuum, ψ_{01} and ψ_{02} are the surface potential of the spheres, κ is given by Eq. (3.2-37), k_B is Boltzmann's constant, e is the electronic charge, T is the temperature, z_i is valency, and $\rho\infty_i$ is the number density of ions in the reference solution where the potential is taken to be zero and the Debye length $1/\kappa$ is the characteristic "thickness" of diffuse electric double layer.

Direct measurement experimental results have shown that constant surface charge boundary conditions describe the electrical double-layer interaction much better than the constant surface potential boundary condition at small separation between the two surfaces.

3.2.8 Photophoresis

The photophoretic force F_i^{Phot} is caused by the presence of light incident on a particle. Two different interactions are part of this mechanism: radiometric effects and radiation pressure.

$$F_i^{\mathrm{Phot}} = F_i^{rm} + F_i^{rp} \tag{3.2-46}$$

The following two sections discuss the various forces on a particle caused by light.

3.2.8.1 Radiometry

Radiometry refers to cases when an illuminating light on a particle creates a temperature gradient in the particle, which in turn creates a temperature gradient in the surrounding fluid. The temperature gradient in the fluid produces a special case of thermophoresis. The distinction between a photophoretic radiometric force and a thermophoretic force is that in the first the temperature gradient originates from within the particle and in the second originates in the surrounding fluid. Yalamov et al. [20] gave the following expression for the radiometric force on a particle when $\mathrm{Kn} < 1$:

$$F_i^{rm} = -\frac{2\pi d v^2 J_i K I}{\rho_f T_s k_p} \tag{3.2-47}$$

where J_i is a measure of the asymmetry of the internal heat source, K is the thermal absorption factor, I is the incident light intensity, T_s is the surface temperature of the illuminated particle, d is the particle diameter, ρ_f is the density, and k_p is the thermal conductivity of the particle.

3.2.8.2 Radiation Pressure

The second photophoretic interaction between a particle and an illuminating light beam is called radiation pressure. This term is a result of the direct transfer of momentum to the particle by the deflection and absorption of light. Any change in the momentum of the incoming photons must be balanced by an equal and opposite particle momentum. Hinds [17] gave the following equation for the radiation pressure force on a particle:

$$F_i^{rp} = -\frac{I\pi D^2 Q_i^{rp}}{4c} \tag{3.2-48}$$

where c is the velocity of light and Q_i^{rp} is the fraction of light geometrically incident on the particle that is effective in transferring momentum to the particle in the direction away from the light source. Q_i^{rp} is a complicated function of particle size, wavelength, index of refraction, and absorption coefficient. Values are not generally available for particles of various materials. The magnitude of the radiation pressure force is predicted to be much smaller than the radiometric force. Only with highly focused, high-power lasers will the light intensity be great enough to produce forces comparable with the gravitational force.

3.2.9 Turbulence Effect

Turbulence will affect the particle motion by modifying the fluid velocity near the particle. In evaluating the trajectory of a particle in a turbulent flow field, the instantaneous component of the fluid velocity must be specified. While the mean velocity and the mean fluctuation kinetic energy may be obtained from a variety of conventional turbulence models, the instantaneous fluctuating velocity components are usually unknown. These fluctuations are random fluctuations in space and time, which are nearly Gaussian processes for homogeneous flows. Monte Carlo velocity simulation techniques have been used successfully as a method for generating time histories that have the random characteristics and statistical properties of turbulence.

The fluctuation kinetic energy κ is given as:

$$\kappa = \frac{1}{2}\sum_{i=1}^{3}\overline{u_i'^2} = \frac{3}{2}\overline{u'^2} \tag{3.2-49}$$

where

$$\overline{u_1'^2} = \overline{u_2'^2} = \overline{u_3'^2} = \overline{u'^2} \tag{3.2-50}$$

In this equation a prime quantity stands for the fluctuation component and a bar on top of a letter denotes the average expected value.

3.2.10 DLVO Theory

The DLVO theory was developed in 1940s by Derjaguin, Landau, Verwey, and Overbeek [21,22]. The theory describes the force between surfaces interacting through a liquid medium. It combines the effects of the van der Waals attraction and the repulsion or attraction as a result of the so-called double-layer of counter-ions previously discussed in Section 3.2-7. The total net interaction between two surfaces, including van der Waals interaction and electrical double-layer interaction, forms the basis of the DLVO theory of colloidal stability.

3.3 PARTICLE ADHESION

3.3.1 van der Waals Force

In the absence of an external force on a particle, particle adhesion to a surface is governed by the omnipresent van der Waals force. For all atoms and molecules, even for nonpolar ones, there exist instantaneous dipoles, caused by the interactions of the electrons surrounding an atom's nucleus. The interaction between these dipoles and the induced dipoles in neighboring atoms results in the dispersion forces of atoms or molecules, which is the dominant portion of the van der Waals force.

Hamaker [23] calculated the interaction between macroscopic bodies, simply by using the additive principle of interactions between all molecules in each body. As a result, the interaction force, here defined as the van der Waals force, between a sphere of radius R and a flat plate at a separation distance d_0 is given by:

$$F_{\mathrm{vdW}} = \frac{AR}{6\,d_0^2} \tag{3.3-51}$$

where A is the conventional Hamaker constant. The relations between the Hamaker constants of two dissimilar materials [24] may be represented by:

$$A_{12} = \sqrt{A_{11}A_{22}} \tag{3.3-52}$$

where A_{11} and A_{22} are the Hamaker constants for materials "1" and "2". In the presence of a medium denoted by "3", the net interaction between materials 1 and 2 is given by:

$$A_{132} = \left(\sqrt{A_{11}} - \sqrt{A_{33}}\right)\left(\sqrt{A_{22}} - \sqrt{A_{33}}\right) \tag{3.3-53}$$

where A_{33} is the Hamaker constant for the medium "3". The Hamaker constant is an intrinsic property of materials. Table 3.3-1 shows the Hamaker constants of materials of interest for semiconductor processing. The greater the Hamaker constant is, the larger the resulting van der Waals attractive force.

3.3.2 Electrostatic Forces

The sign and magnitude of the zeta potential on the surface determines the electrostatic force. If the zeta potentials of a particle and a surface have the same positive or negative charge, the electrostatic force is defined as repulsive, otherwise the force is attractive. The pH and ionic strength of a solution affect the magnitude of the zeta potential and double-layer thickness. As the solution's pH increases, the zeta potentials of the surface move to a more negative value because of the dominant adsorption of OH^-in alkaline pH solutions.

TABLE 3.3-1 Hamaker Constants of Materials of Interest in Semiconductor Processes [19,24,27]

Materials	Hamaker Constant ($\times 10^{-20}$ J)
Si	25.6
SiO_2	6.5
Al_2O_3	15.5
PSL (polystyrene latex)	6.5
Si_3N_4	17
TaN	28.03
H_2O	3.7
Organic materials	2–10

Compiled by the authors from references.

3.3.2.1 *Effect of pH*

In an aqueous media, the pH of the medium of the particle and substrate are important factors that affect the zeta potential. Eq. (3.2-44) shows that the particle charge is proportional to the zeta potential. Total surface charge, σ_{surf} is calculated by the following equation:

$$\sigma_{surf} = zF(\Gamma_{H^+} - \Gamma_{OH^-}) \tag{3.3-54}$$

where z is the valance, F is Faraday constant, and Γ is the adsorption density of H^+ and OH^-. This equation indicates that pH directly determines the sign of zeta potentials. Fig. 3.3-4 shows the zeta potentials of materials of interest to semiconductor processes as a function of pH.

Generally, the zeta potential has a positive value at acidic pH because of the preponderance of H^+ ions in the acidic region at the surface of the material. The zeta potential reduces to zero and then becomes negative as the pH increases to an alkaline value. However, this tendency can vary for various materials. The zero point of the zeta potential is known as the isoelectric point (IEP). At IEP, the net electrophoretic mobility of particles is zero. Table 3.3-2 summarizes the IEP of some metal oxides.

3.3.2.2 *Effect of Ionic Strength*

The ionic strength of a solution, I, is the concentration of ions in the solution:

$$I = \frac{1}{2}\sum_{i=1}^{n} C_i z_i^2 \tag{3.3-55}$$

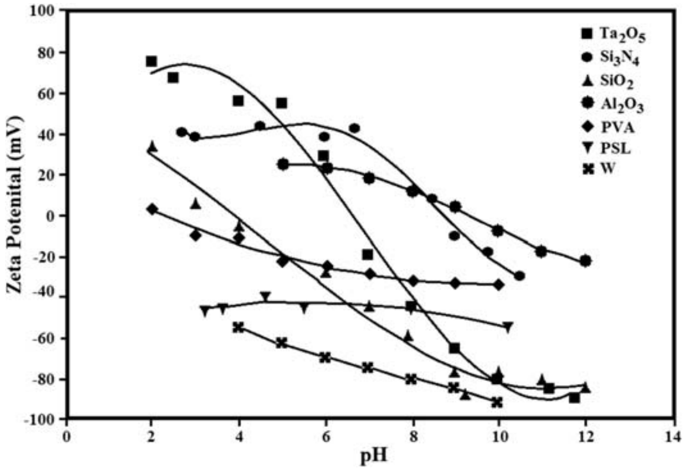

FIGURE 3.3-4 Zeta potential of materials as a function of pH. *Used with permission of the authors.*

TABLE 3.3-2 Isoelectric Point of Materials of Interest in Semiconductor Processes

Materials	Isoelectric Point
Si	3–4
SiO_2	1.5–3.7
Al_2O_3	7.4–9.5
Si_3N_4	3–5.5
TiO_2	6.0–6.7
PVA	−2
CeO_2	6–8

PVA, polyvinyl alcohol.
Compiled by the authors from references.

where C is the molar concentration and z is the charge of each ion, i. From Eqs. (3.2-35) and (3.2-37), κ is proportional to $\sqrt{I}$, the square root of the ionic concentration of solution. Fig. 3.3-5 shows the influence of electrolyte concentration, κ, on total interaction forces because of the double-layer compression at higher concentration, which results in a more attractive electrostatic force.

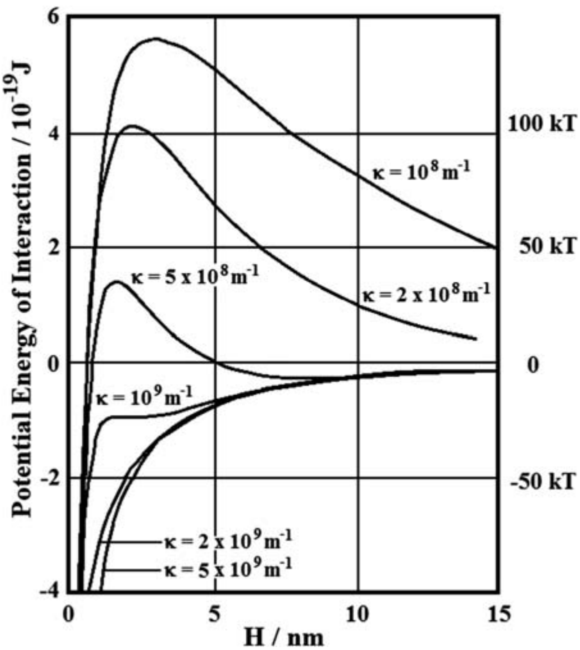

FIGURE 3.3-5 The effect of electrolyte concentration, κ, on total interaction energy [38]. *Reprinted from D.J. Shaw, B. Costello, Introduction to Colloid and Surface Chemistry, fourth ed., Butterworth-Heinemann, Oxford, 1993. Copyright (1993), with permission from Elsevier and Butterworth-Heinemann.*

3.3.3 Short-Range Forces

The primary difference between long-range and short-range forces is that short-range forces act exclusively within the contact zone, whereas long-range forces act both within and outside the region of contact. Chemical bonding is an example of short-range forces. A chemical bond is the physical phenomenon of chemical substances being held together by attraction of atoms to each other through sharing of electrons and exchanging of electrons. In general, strong chemical bonds are found in molecules, crystals, or solid metals where the atoms are organized in ordered structures. Weak chemical bonds are explained classically to be effects of polarity, or the lack of strong bonds.

3.3.4 Capillary Condensation

When moisture is present in the air, condensation can take place between a particle and a substrate, as shown in Fig. 3.3-6. The meniscus that is formed

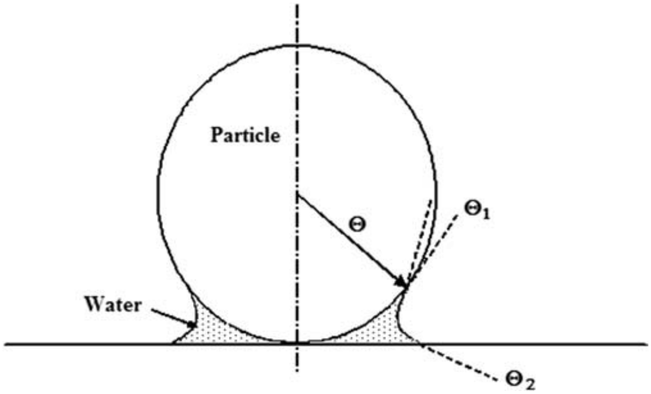

FIGURE 3.3-6 Capillary condensation between a particle and a substrate. θ represents the contact angle. *Used with permission of the authors.*

draws the bodies together as a result of the surface tension force around the circumference F_{ST} and the negative Laplace pressure of the liquid F_P:

$$F_{ST} = 2\pi R\gamma_L \sin\phi \sin(\phi + \theta_1) \tag{3.3-56}$$

where R is radius of the sphere, γ_L is the surface tension of the liquid, and θ is the contact angle. When the filing angle ϕ is small, F_P is given by the following equation.

$$F_P = 2\pi R\gamma_L(\cos\theta_1 + \cos\theta_2) \tag{3.3-57}$$

For the case of small ϕ and equal contact angle on each surface $\theta_1 = \theta_2$, the following simple equation is used:

$$F_{Cap} = F_{ST} + F_P \approx 4\pi R\gamma_L \cos\theta \tag{3.3-58}$$

The measured adhesion force [25] is normally less than the value given by Eq. (3.3-56) and approaches this value when relative humidity approaches 100%.

Fig. 3.3-7 shows the capillary force for a 0.1-μm-diameter particle as a function of surface tension. Isopropyl alcohol and deionized water (DI H_2O) have surface tensions of 22 and 72 dyn/cm, respectively. If the surface tension increases, the capillary force also linearly increases.

Fig. 3.3-8 shows a schematic diagram of a particle adhering to the surface (1) by only van der Waals force and (2) by van der Waals and capillary forces. The distance (d_0) between a particle and the surface is generally 4 Å. When the capillary force affects particle adhesion, a liquid medium exists between the particle and the surface.

In DI H_2O, the van der Waals force of alumina (Al_2O_3) and silica (SiO_2) particles on the flat surface and the capillary force are shown in Fig. 3.3-9.

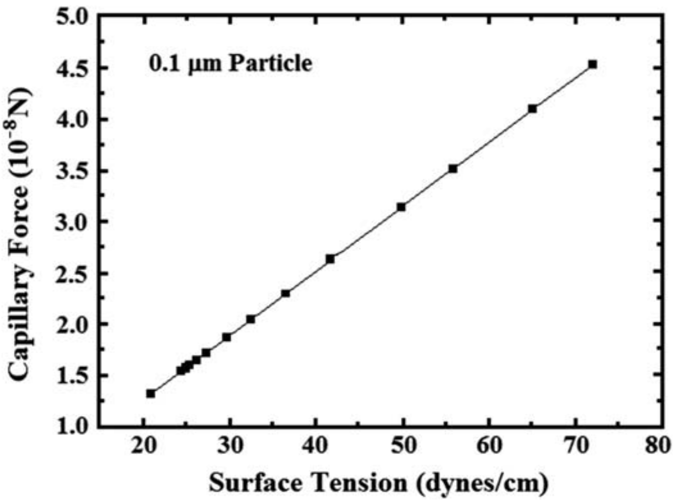

FIGURE 3.3-7 The capillary force as a function of the surface tension of liquid. *Used with permission of the authors.*

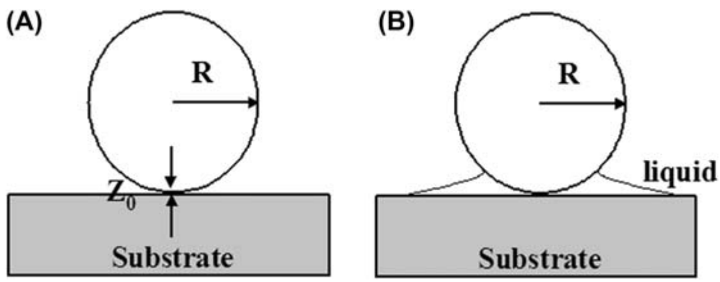

FIGURE 3.3-8 Schematic diagram of a particle adhering to the surface (A) by only van der Waals force and (B) by van der Waals and capillary forces. Z_0 is the distance of separation between the particle and surface. *Used with permission of the authors.*

The van der Waals forces of these two particles of different materials are different because of the differences in their Hamaker constants. However, the capillary force of the particles of different materials, which are the same size, are equal because the capillary force is affected only by the surface tension of the liquid media, the surface energy of the particle and substrate, and the particle size. This indicates that the material physical properties of the particle and the surface do not influence the capillary force provided they have the same surface energy. The capillary force is a much more dominant force for particle adhesion compared with van der Waals and electrostatic forces.

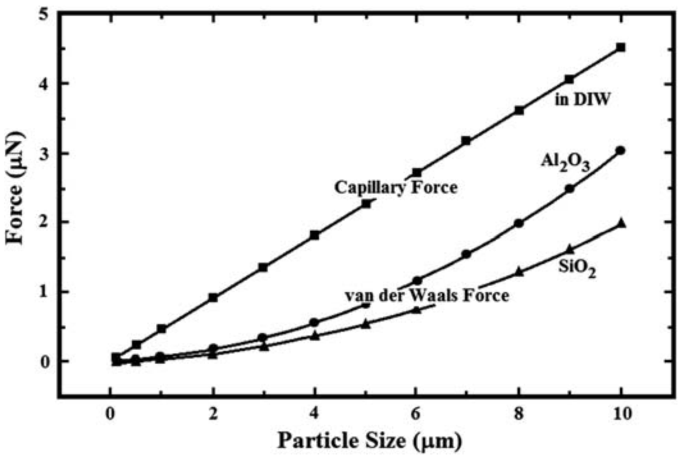

FIGURE 3.3-9 The capillary and van der Waals forces as a function of particle size. *Used with permission of the authors.*

3.3.5 Effect of Deformation on Adhesion

Deformation of particles makes particle removal more challenging. The van der Waals force, F_{vdW}, causes deformation when a particle is softer than a substrate or vice versa [26]. The increase in the contact area (at the interface) because of the deformation increases the van der Waals force. Eq. (3.2-58), for the measured adhesion force, does not take into account the additional adhesion force caused by deformation; however, Bowling [25] gave the total adhesion force including the component caused by the deformation as:

$$F_{ad} = F_{vdW} + F_{vdW-deform} = \frac{AR}{6d_0^2}\left(1 + \frac{a^2}{Rd_0}\right) \tag{3.3-59}$$

where A is the Hamaker constant, R is the radius of the spherical particle, d_0 is the separation distance between the particle and the substrate, and a is the contact radius between the deformed particle and the surface. For smooth surfaces, d_0 is taken as 4 Å.

Fig. 3.3-10 shows the flattening of a PSL particle and subsequent increase of the contact area on a silicon wafer with native oxide [29]. Another important factor affecting the deformation of particle is its time dependency. Fig. 3.3-11 shows that the contact area increases as time increases [28,29].

Understanding of deformation caused by particle adhesion is dominated by three theories: DMT (Derjaguin–Müller–Toporov) [26], JKR (Johnson–Kendall–Roberts) [30], and MP (Maugis and Pollock) models [31]. The JKR and DMT models are used to describe adhesion of elastically deforming system. The MP model accounted for tensile interactions and plastic deformations.

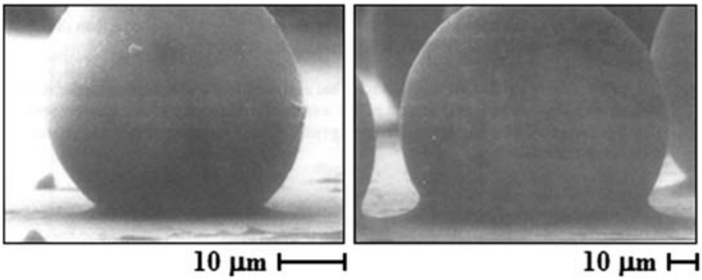

FIGURE 3.3-10 The change of cantilever shape as a function of applied force on a cantilever. *Used with permission of the authors.*

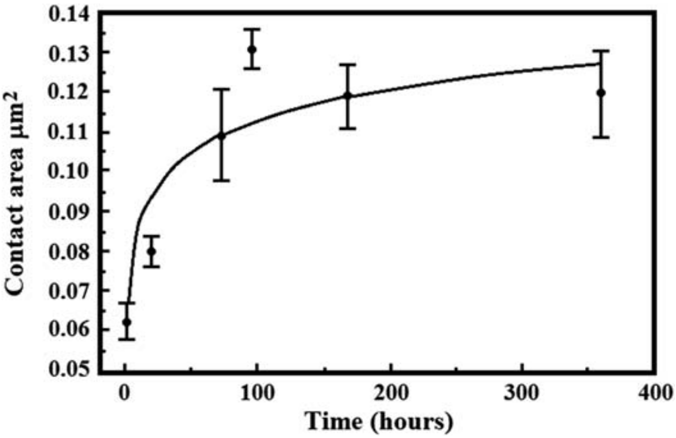

FIGURE 3.3-11 Measured adhesion force of wafer surface at various pH ranges [35]. *Reproduced by permission of ECS—The Electrochemical Society from S.Y. Lee, S.H. Lee, J.G. Park, J. Electrochem. Soc.*

Both JKR and DMT have their ranges of validity. The range of each theory is defined in terms of a dimensionless parameter μ [26,30]:

$$\mu = \frac{32}{3\pi}\left[\frac{2Rw_A^2}{\pi E^{*2} d_0^3}\right]^{1/3} \tag{3.3-60}$$

where w_A is work of adhesion and E^* is the reduced Young's modulus given by:

$$1/E^* = \left(1 - v_1^2\right)/E_1 + \left(1 - v_2^2\right)/E_2 \tag{3.3-61}$$

where, E_1 and E_2 are elastic modulus and v_1 and v_2 are Poisson ratios associated with each body.

According to the so-called MYD model, mobility values of $\mu > 1$, which indicate higher surface energy materials and relatively large particles, correspond to JKR theory. According to JKR:

$$a = \left(\frac{6\pi w_A R^2}{K}\right)^{1/3} \tag{3.3-62}$$

where, a is the contact area radius. Similarly, $\mu < 1$ corresponds to the DMT model, which indicates lower surface energy and smaller particles. According to DMT [32],

$$a = \left(\frac{\pi w_A R^2}{K}\right)^{1/3} \tag{3.3-63}$$

For both these equations, K is calculated as follows:

$$K = 4E^*/3 \tag{3.3-64}$$

The actual value of the contact radius predicted by the JKR theory is approximately twice that predicted by the DMT model; therefore, it is necessary to establish which theory correctly describes a system.

Maugis and Pollock expanded the JKR model to the region where the surface force—induced stress exceeds the yield strength of the contacting material [33], which is the plastic region:

$$a = \left(\frac{2w_A R}{3Y}\right)^{1/2} \tag{3.3-65}$$

where w_A is work of adhesion between the particle and surface and Y is the yield strength of the softer material. If the two contact materials are in a third medium such as H_2O, the work of adhesion is given by:

$$w_A = w_1 + w_2 + w_{12} \tag{3.3-66}$$

where the w_1 and w_2 are the adhesive energies of the two surfaces and w_{12} is an interaction term.

3.3.6 Particle Adhesion Force Measurements

The particle adhesion forces in air and liquid media are different because the adhesion forces are different in these different media. The main adhesion force in air is van der Waals forces, whereas the adhesion force of a particle in a liquid media is calculated by the sum of the electrostatic force and van der Waals force. The van der Waals force is always an attractive force, but the electrostatic force is affected by surface potentials such as zeta potential and can be attractive or repulsive. If the zeta potential of a particle and a surface has the same positive or negative charge, the electrostatic force is repulsive.

Therefore, particle adhesion and contamination occur when the electrostatic force is attractive and the attractive van der Waals force is larger than the repulsive electrostatic force. Even though the DLVO theory-based calculation takes into account only the Hamaker constant, zeta potentials, and particle size in a liquid medium, it nevertheless provides a good estimate for the particle adhesion force. Experimental measurement of particle adhesion forces on a surface can provide an excellent way to evaluate the effect of chemistry during cleaning and chemical mechanical polishing processes on particle adhesion.

3.3.6.1 Adhesion Force Measurement by Atomic Force Microscopy

Atomic force microscopy (AFM) has been used to study phenomena such as abrasion, adhesion, cleaning, corrosion, etching, friction, lubrication, polishing, and force of adhesion of particles [34]. AFM not only images the surface with resolution on the atomic scale but also measures the interaction forces between the cantilever tip and the surface at the nanometer scale. The AFM observes the surface of a sample using a probe with a sharp tip, approximately 2 μm long and often less than 10 nm in diameter. The tip is placed at the free end of a cantilever that is 100−200 μm. The force between the tip and the surface causes the cantilever to bend or deflect. This bending is quantified into length and a map of the surface is obtained showing the peaks and valleys.

Instead of using a tip of cantilever, a particle, which is attached on a tipless cantilever, can be used to measure the interaction forces between the particle and surface. The force versus distance (F vs. d) curves [34,35] are used to measure the vertical force applied to the tip while in contact with the surface. Local variations in the form of the F versus d curve indicate variations in the local elastic properties. The force curves show the deflection of the free end of the cantilever as the fixed end of the cantilever is brought vertically toward and then away from the sample surface. The cantilever positions at several points along the force curve are shown in Fig. 3.3-12.

Point (a) in Fig. 3.3-12, shows the initial cantilever position, the approach, when the tip does not touch the surface. This is considered the control point. In this region, if the cantilever feels a long-range attractive or repulsive force, it will deflect downward or upward before making contact with the surface. In the case as shown in Fig. 3.3-12, point (a), there is minimal long-range force, so the noncontact part of the force curve shows no deflection. Point (b) shows that the probe tip is very close to contacting the surface and may jump into contact if sufficient attractive force is felt from the sample. Point (c) shows the tip in contact with the surface, and thus, the cantilever deflection will increase as the fixed end of the cantilever is brought closer to the sample. If the cantilever is sufficiently stiff and the material is soft, the probe tip may indent into the surface at this point. In this case, the slope or shape of the contact part of the force curve provides information about the elasticity of the sample surface. Point (d) shows that after loading the cantilever to a desired force

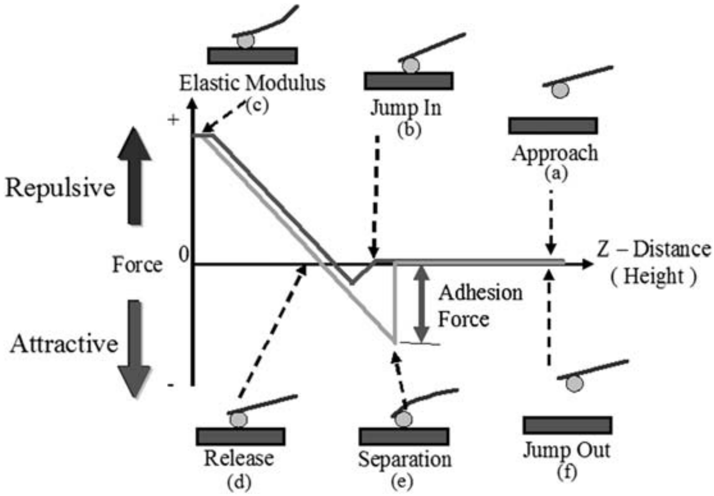

FIGURE 3.3-12 The zeta potential of wafer surfaces as a function of pH [35]. *Reproduced by permission of ECS—The Electrochemical Society from S.Y. Lee, S.H. Lee, J.G. Park, J. Electrochem. Soc.*

value, the process is reversed. As the cantilever is withdrawn, adhesion formed during contact with the surface may also cause the cantilever to adhere to the sample some distance past the initial contact point on the approach curve (b). A key measurement of the AFM force curve is the point (e) at which the adhesion is broken and the cantilever lifts from the surface, known as the jump-out point. This method can be used to measure the separation force required to break the bond or adhesion of the particle [36].

The force curve, shown in Fig. 3.3-12, shows the relationship between the control point and the deflection of the cantilever. Because the set point defines the value of the deflection signal maintained by the feedback loop, the force curve can be used to calculate the nominal contact force of the tip on the sample if the spring constant, k, of the cantilever is known. The adhesion force of the particle is defined by the equation:

$$F = -k\Delta d \tag{3.3-67}$$

where, Δd is the distance from the control point to the jump-out point.

3.3.6.2 Measured Adhesion Forces in Liquid Media

The measured adhesion forces of a particle on a thin-film surface in solutions of acidic, neutral, and basic pH are shown in Fig. 3.3-13 [35]. To compare the adhesion force of various substrates, colloidal silica particles are used as references. The attractive forces of silica particles on SiLK, TEOS (tetraethylorthosilicate), Cu, and TaN wafers are larger in acidic solutions. As shown

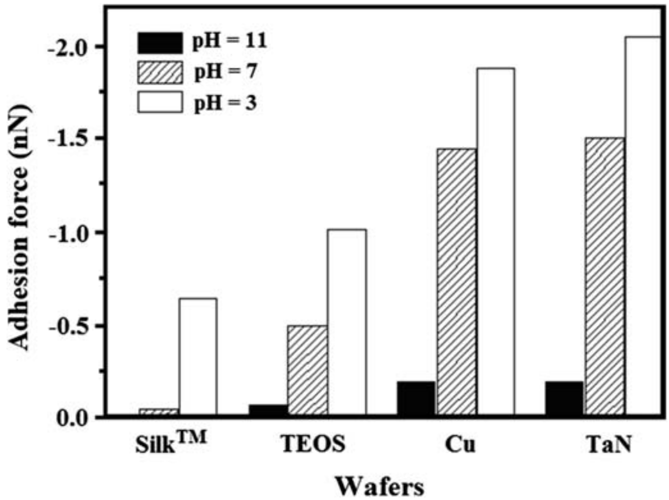

FIGURE 3.3-13 The adhesion forces between Cu wafers and spherical silica particle in different solutions [36]. BTA is benzotriazole. *Reproduced by permission of ECS—The Electrochemical Society from Y.K. Hong, D.H. Eom, S.H. Lee, T.G. Kim, J.G. Park, A.A. Busnaina, J. Electrochem. Soc.*

in Fig. 3.3-14, the repulsive electrostatic force is smaller in acidic than in alkaline solution because of the decrease in zeta potentials of the thin films on the wafers as the pH decreases. This also shows that the particle adhesion is accelerated under low-pH conditions. The measured interaction forces show the same trends as the calculated interaction forces over the range of pH values considered. Adhesion of particles on surfaces in acidic solution was stronger than that in alkaline solution. Tantalum nitride surfaces show the strongest adhesion of silica particles in the pH range investigated. The lowest adhesion force of particles on the tested substrates was measured on a low−dielectric constant (low-*k*) organic polymer, Dow SiLK. The stronger repulsive forces between silica particles and SiLK surfaces are attributed to double-layer electrostatic forces [35].

The particle adhesion forces of silica particles on Cu wafers in three different cleaning solutions were compared with those in DI H_2O [36]. The lowest adhesion force was observed to be 0.0124 nN in citric acid ($C_2H_6O_4$) with NH_4OH at $pH = 6$. On the other hand, the highest adhesion force of 8.87 nN was measured in the citric acid solution with tetramethyl ammonium hydroxide (TMAH), also with a $pH = 6$. The adhesion force in the solution containing NH_4OH was two orders of magnitude lower than that in the solution with TMAH, as shown in Fig. 3.3-15. These results show that the appropriate selection of the pH and the chemical additives are very important in the design of the cleaning solution to achieve optimum particle removal efficiency.

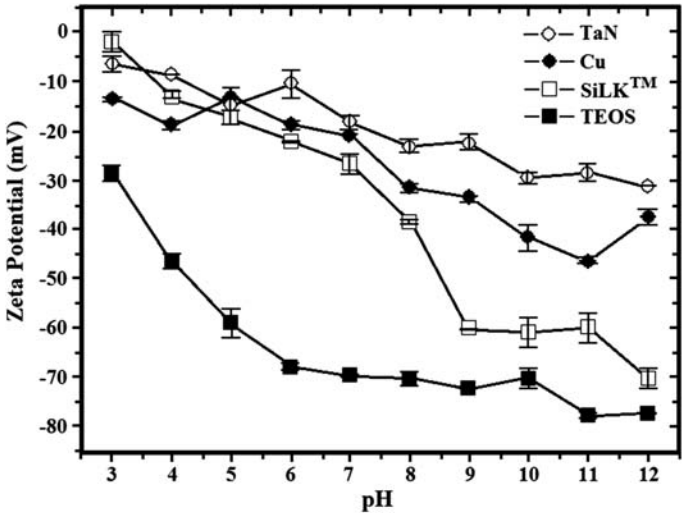

FIGURE 3.3-14 Deformation of PSL particles over time showing flattening of the shape because of van der Waals forces [29]. *Reprinted from Koinklijke N.V. Brill, Fundamentals of Adhesion and Interfaces, 1995. Used with permission.*

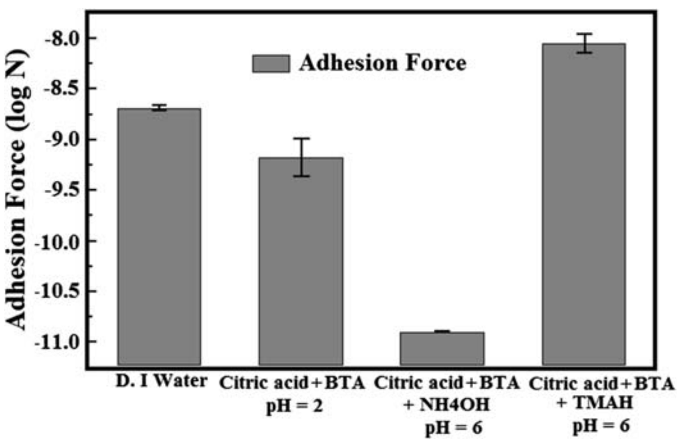

FIGURE 3.3-15 Deformation of PSL particles over time showing change in the contact area because of van der Waals forces [28]. *Reprinted from Koninklijke N.V. Brill, J. Adhesion Sci. Technol., 1994. Used with permission.*

3.4 PARTICLE REMOVAL

Fabrication of microelectronics requires nanoscale particle removal along with other contaminant removal from the wafer surface, independent of the top thin films. These undesired particles influence the device yield and reliability. Various cleaning methods have been developed to remove particles from the substrate. For semiconductor wafer cleaning, brush scrubbing, laser shock wave, supercritical CO_2 cleaning, and megasonic cleaning are widely used in the semiconductor industry. Particle removal techniques are selected based on their effectiveness and their damage potential to features on the wafer or lack of it.

As the size of structures shrinks with each new generation of devices, it becomes more difficult to remove nanoscale particles from patterned wafers. For many of the cleaning methods, there is a tradeoff between cleaning and damage. One of the effective methods for nanoparticle removal is megasonic cleaning. However, nanostructures are vulnerable to damage when this method is used. The mechanism of damage and the control parameters that enable damage-free removal of nanoparticles are described in the next subsection.

The damage-free particle removal occurs by one of the three mechanisms of action below:
Lifting: removal occurs if the Saffman lift force is greater than force of adhesion $F_1 \gg F_a$
Sliding: removal occurs by the hydrodynamic drag force when it exceeds the resultant normal forces on the particle $F_d > k\ (F_a - F_1)$, where k is the coefficient of friction
Rolling: removal occurs when the moment of drag force about the particle contact point O at area a exceeds that exerted by the resultant normal force and is given as $F_d\ (r - \alpha_0) > (F_a - F_1)$ [37].

3.4.1 Megasonic Damage Mechanism

In ultrasonic and megasonic tanks, the transducer produces waves in the fluid by oscillating in reference to an electrical signal at a given frequency. This creates compression waves in the liquid of the tank, which at lower frequencies tear the liquid apart, leaving behind many millions of microscopic voids or vacuum bubbles (cavitation). The existence of cavitation at low ultrasonic frequencies, up to 100 kHz, is well known. However, at higher megasonic frequencies, there is insufficient time between pulses to allow the formation of cavitation bubbles.

It is this formation and collapse of the bubbles of either gas or vapor in a liquid that is associated with the damage to nanostructures. Measurements have shown that for a traditional megasonic tank, although the tank is operating at high frequency (e.g., 760 kHz), the transducers are generating high amplitude at low frequencies (40,200 kHz) as well. Therefore, it has been experimentally verified that cavitation damage occurs by secondary frequencies as low as 40 KHz in the megasonic tanks. It is the existence of these high-powered low frequencies in megasonic tanks that generate ultrasonic cavitation responsible for damage.

3.4.2 Damage-Free Megasonic Particle Removal

As discussed, traditional megasonic tank transducers generate many frequencies as low as 40 kHz at high amplitude (power). Excluding these low frequencies using a narrow band transducer will eliminate damage even at high power.

Polysilicon line structures on 200-mm wafers, with the smallest line width of 70 nm, are used for these damage tests. It has been showed that all significant variables such as the power of the megasonic sound field, the amount of dissolved gas, the size of gas bubbles, the process temperature, and the ratio of the ammonia hydroxide hydrogen peroxide mixture have influence on the process results. One of the most significant influences on the cleaning and damage process happen because of the bubble distribution (the amount of bubbles in the active megasonic field) and the size distribution of stimulated gas bubbles.

The results shown in Fig. 3.4-16 are one of the many test results confirming that using traditional megasonic tank power setup, polysilicon structures have been damaged while samples cleaned using the narrow-bandwidth megasonic tank did not show any damage. By eliminating low frequencies, damage-free

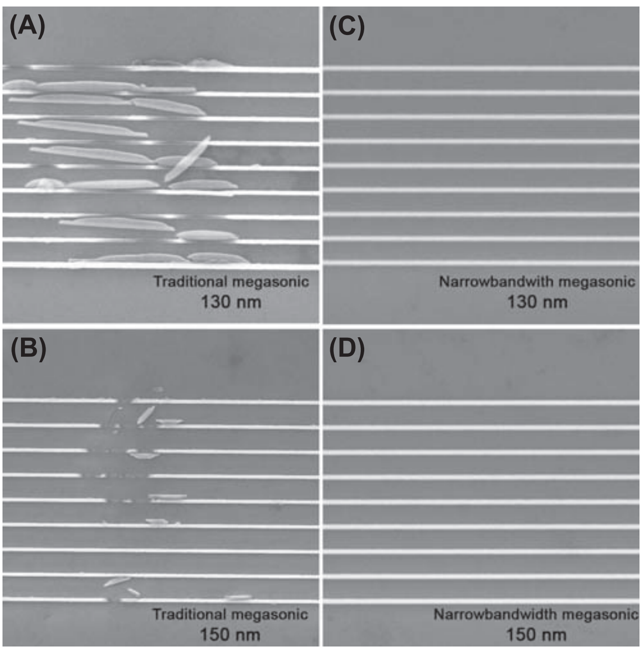

FIGURE 3.4-16 Scanning electron microscope images of 130-nm (A and C) and 150-nm (B and D) lines after cleaning with 50% power for 5 min. While the single-wafer megasonic tank damages the structures, the narrow bandwidth transducer preserves the patterns. *Used with permission of the authors.*

removal of nanoscale particles was demonstrated at high amplitude (power) without sacrificing effective cleaning.

3.5 SUMMARY

In this chapter, forces acting on particles during transport, deposition, and adhesion are reviewed. Most contaminant particles are generated during wafer processing in integrated circuit manufacturing. Particulate contamination needs to be removed before each manufacturing processor step to avoid low device yield and boost reliability. These generated particles become suspended in a medium (gas or liquid) that will provide the means for their transport and deposition. Drag force, electrophoretic force, Brownian diffusion, thermophoretic force, etc. are some of the forces covered in this chapter that could directly affect particle transport and deposition. After a particle deposits on a substrate, it becomes governed by the adhesion force. The adhesion force is dominated by van der Waals force, which could increase if particle or substrate deformation occurs. In liquids, the particle—substrate interaction will experience an additional force — the double-layer electrostatic force which depends on the pH and ionic strength of the solution. In addition, the capillary force could also be dominant if moisture exists at the interface between a particle and substrate in a gas medium. Adhered particles are removed using many cleaning technologies such as brush and megasonic cleaning, which are two of the most widely used particle removal techniques in the semiconductor industry.

REFERENCES

[1] N.A. Fuchs, The Mechanics of Aerosols, Macmillan, New York, 1964.
[2] M.E. O'Neill, Chem. Eng. Sci. 23 (11) (1968) 1293.
[3] P.G. Saffman, J. Fluid Mech. 22 (2) (1965) 385.
[4] J.B. McLaughlin, J. Fluid Mech. 224 (1991) 261.
[5] R.A. Serway, J.W. Jewett, Physics for Scientists and Engineers with Modern Physics, Thomson Brooks/Cole, Stamford, CT, 2004.
[6] N. von Smoluchowski, in: J.A. Barth (Ed.), Handbuchder Electrizität and des Magnetismus, vol. 2, 1921, p. 366. Leipzig, Germany.
[7] E. Hückel, Phys. Zeitsch Ger. 25 (1924) 204.
[8] R. Brown, Phil. Mag. 4 (1828) 161.
[9] A. Li, G. Ahmadi, Aerosol Sci. Technol. 16 (4) (1992) 209.
[10] A. Einstein, Investigation on the Theory of Brownian Movement, Dover, New York, 1956.
[11] P.W. Smith, A. Ashkin, W.J. Tomlinson, Opt. Lett. 6 (6) (1981) 284.
[12] J. Tyndall, On dust and disease, Proc. R. Inst. Gt. Br. 6 (1870) 1.
[13] P.S. Epstein, Z. Phys. 54 (7—8) (1929) 537.
[14] J.R. Brock, J. Colloid Sci. 17 (8) (1962) 768.
[15] L. Waldmann, K.H. Schmitt, in: C.N. Davies (Ed.), Aerosol Science, Academic Press, New York, 1966.
[16] G.K. Batchelor, C. Shen, J. Colloid Interface Sci. 107 (1) (September 1985) 21—37.

[17] W.C. Hinds, Aerosol Technology, Wiley-Interscience Publication, New York, 1982.
[18] R.B. Bird, W.E. Stewart, E.N. Lightfoot, Transport Phenomena, Revised second ed., John Wiley & Sons, Hoboken, NJ, 2006.
[19] R. Hogg, T.W. Healy, D.W. Fuerstenau, Trans. Faraday Soc. 62 (1966) 1638.
[20] Y.I. Yalamov, V.B. Kutukov, E.R. Shchukin, J. Colloid Interface Sci. 57 (3) (1976) 564.
[21] B.V. Derjaguin, L. Landau, Acta Physiochem. 14 (1941) 633.
[22] E.J.W. Verwey, J.Th G. Overbeek, Theory of the Stability of Lyophobic Colloids, Elsevier, Amsterdam, 1948.
[23] H.C. Hamaker, Physica 4 (10) (1937) 1058.
[24] J. Visser, in: E. Matijevic (Ed.), Surface and Colloid Science, vol. 8, John Wiley and Sons, New York, 1976.
[25] R.A. Bowling, in: K.L. Mittal (Ed.), Particles on Surfaces I: Detection, Adhesion and Removal, Plenum Press, New York, 1988, p. 129.
[26] B.V. Derjaguin, V.M. Müller, Y.P. Toporov, J. Colloid Interface Sci. 53 (2) (1975) 314.
[27] P.C. Hiemenz, Principles of Colloid and Surface Chemistry, Marcel Dekker, New York, 1986, p. 645.
[28] S. Krishnan, A.A. Busnaina, D.S. Rimai, D.P. DeMejo, J. Adhesion Sci. Technol. 8 (11) (1994) 1357.
[29] S. Krishnan, A.A. Busnaina, D.S. Rimai, L.P. DeMejo, in: D.S. Rimai, L.P. DeMejo, K.L. Mittal (Eds.), Fundamentals of Adhesion and Interfaces, VSP BV press, The Netherlands, 1995, p. 95.
[30] K.L. Johnson, K. Kendall, A.D. Roberts, Proc. R. Soc. Lond. Ser. A 324 (1558) (1971) 301.
[31] D. Maugis, H.M. Pollock, Acta. Metall. 32 (9) (1984) 1323.
[32] A.M. Freitas, M.M. Sharma, J. Colloid Interface Sci. 233 (1) (2001) 73.
[33] W.A. Ducker, T.J. Senden, R.M. Pashley, Langmuir 8 (7) (1992) 1831.
[34] J. Drelich, G.W. Tormoen, E.R. Beach, J. Colloid Interface Sci. 280 (2) (2004) 484.
[35] S.Y. Lee, S.H. Lee, J.G. Park, J. Electrochem. Soc. 150 (5) (2003) G327.
[36] Y.K. Hong, D.H. Eom, S.H. Lee, T.G. Kim, J.G. Park, A.A. Busnaina, J. Electrochem. Soc. 151 (11) (2004) G756.
[37] L.-O. Heim, J. Blum, M. Preuss, H.-J. Butt, Phys. Rev. Lett. 83 (6) (1999) 3328.
[38] D. Shaw, J.B. Costello, Introduction to Colloid and Surface Chemistry, fourth ed., Butterworth-Heinemann, Oxford, 1993.